



AQ2.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1:

1 point

Consider a square matrix $A = \begin{bmatrix} -12 & 0 \\ 2 & 0 \\ -10 & 1 \end{bmatrix}$. If

- $P = -1M_{11} - 2M_{12} + 0M_{13}$ $P = -1M_{11} - 2M_{12} + 0M_{13}$
- $Q = -2M_{21} + 0M_{22} + 1M_{23}$ $Q = -2M_{21} + 0M_{22} + 1M_{23}$
- $R = -1M_{31} - 0M_{32} + 1M_{33}$ $R = -1M_{31} - 0M_{32} + 1M_{33}$

where M_{ij} is the minor with respect to the (i, j) -th entry, then choose the set of correct options.

☐

$$\det(A) = -2\det(A) = -2.$$

☐

$$P \neq QP = Q.$$

☐

$$P \neq Q \neq RP = Q = R.$$

☐

$$P = Q = RP = Q = R.$$

☐

$$Q \neq RQ = R$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A) = -2\det(A) = -2.$$

$$P = Q = RP = Q = R.$$

1 point

Suppose for a real $3 \times 3 \times 3$ matrix AA , there exists a real $3 \times 3 \times 3$ matrix PP such that $D = PAP^{-1}$ $D = PAP^{-1}$ is a real $3 \times 3 \times 3$ diagonal matrix. Choose the correct set of options.

☐

$$\det(A)\det(A) \text{ must be equal to } \det(P)\det(P).$$

☐

$$\det(A)\det(A) \text{ must be equal to } \det(D)\det(D).$$

☐

$$\text{The matrix } AA \text{ must be equal to } DD.$$

☐

If DD is the identity matrix of order 33, then AA must be the identity matrix of order 3

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\det(A)\det(A) \text{ must be equal to } \det(D)\det(D).$$

If DD is the identity matrix of order 33, then AA must be the identity matrix of order 3

1 point

Choose the set of correct options for square matrices of order 33.



☐

If $AB = 0$, then one of the matrices between A and B must have determinant 0.

☐

If $AB = 0$, then both the matrices A and B must have determinant 0.

☐

If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{3}{\det(B)}$.
 $\det A = \det(B)3$.

☐

If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{9}{\det(B)}$.
 $\det A = \det(B)9$.

☐

If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{27}{\det(B)}$.
 $\det A = \det(B)27$.

☐

If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $AB = 0$, then one of the matrices between A and B must have determinant 0.

If $AB = 3I$, where I denotes the identity matrix of order 3, then $\det A = \frac{27}{\det(B)}$.
 $\det A = \det(B)27$.

If $AB = 3I$, where I denotes the identity matrix of order 3, then both A and B must have non-zero determinant.

1 point

Consider a matrix $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$. If $a + b + c$ is divisible by 66 then choose the

set of correct options.

☐

$\det(A)$ is divisible by 55.

☐

$\det(A)$ is divisible by 66.

☐

$\det(A)$ is divisible by 22.

☐

$\det(A)$ is divisible by 33.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A)$ is divisible by 66.

$\det(A)$ is divisible by 22.

$\det(A)$ is divisible by 33.

Let $A = [a_{ij}]$ be a square matrix of order 3, where $a_{ij} = i$. Find $\det(A)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Level 2:

Let $A = [\delta_{ij}]$ be a square matrix of order 33, where $\delta_{ij} = i \times j$. Find $\det(A)$.

No, the answer is incorrect.**Score: 0****Accepted Answers:**

(Type: Numeric) 0

1 point

1 point

Consider the following square matrices:

$$A = \begin{bmatrix} 2013 & 2014 & 2015 \\ 2016 & 2017 & 2022 \\ 2019 & 2020 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2017 & 2022 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 4034 & 4044 \\ 2013 & 2014 & 2015 \\ 2019 & 2020 & 2021 \end{bmatrix}$$

$$A = \begin{bmatrix} 2013 & 2016 & 2019 \\ 2014 & 2017 & 2020 \\ 2015 & 2022 & 2021 \end{bmatrix}, B = \begin{bmatrix} 2016 & 2013 & 2019 \\ 2017 & 2014 & 2020 \\ 2022 & 2015 & 2021 \end{bmatrix} \text{ and } C = \begin{bmatrix} 4032 & 2013 & 2019 \\ 4034 & 2014 & 2020 \\ 4044 & 2015 & 2021 \end{bmatrix}$$

Choose the set of correct options.

☐

$$\det(B) = \det(A) \det(B) = \det(A).$$

☐

$$\det(A) = -\det(B) \det(A) = -\det(B).$$

☐

$$\det(C) \neq -2\det(B) \det(C) = -2\det(B).$$

☐

$$\det(C) = -2\det(A) \det(C) = -2\det(A).$$

No, the answer is incorrect.**Score: 0****Accepted Answers:**

$$\det(A) = -\det(B) \det(A) = -\det(B).$$

$$\det(C) \neq -2\det(B) \det(C) = -2\det(B).$$

$$\det(C) = -2\det(A) \det(C) = -2\det(A).$$

(Based on the below information answer questions 8 and 9).Let A be a real 3×3 matrix as given below:

$$A = [C_1 \ C_2 \ C_3]$$

where C_i represents the i -th column of the matrix A . Now let us consider the following set of matrices.

- $A_1 = [C_1 \ C_2 + 5C_3 \ C_3]$ $A_1 = [C_1 C_2 + 5C_3 C_3]$
- $A_2 = [C_1 + C_2 + C_3 \ C_2 \ C_3]$ $A_2 = [C_1 + C_2 + C_3 C_2 C_3]$
- $A_3 = [C_1 \ C_2 + 5C_3 \ 0]$ $A_3 = [C_1 C_2 + 5C_3 0]$
- $A_4 = [C_1 + C_2 \ C_2 + C_3 \ C_3 + C_1]$ $A_4 = [C_1 + C_2 C_2 + C_3 C_3 + C_1]$

1 point

Which of the matrices have the same determinant as that of A ?

☐

A_1

☐

A_2

☐

A_3

☐

A_4

No, the answer is incorrect.

Score: 0

Accepted Answers:

A_1

A_2

1 point

Choose the set of correct options.

☐

$\det(A_3)$ cannot be determined from the given information.

☐

$\det(A_3) = 0$

☐

$\det(A_4) = 2 \det(A)$

☐

$\det(A_2) = 3 \det(A)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\det(A_3) = 0$

$\det(A_4) = 2 \det(A)$

If the diagonal entries of a 3×3 lower triangular matrix A are 1, 2, and 3, then find the sum of roots of the equation $\det(A - xI) = 0$, where I is the identity matrix of order 3.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 6

1 point

[Check Answers](#)

Your score is: 0/10